Hub Operations Digital Twin — v2.1.1 CURE×ZIP Coupling

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Version: 2.1.1 — One increment toward v2.2. Introduces the CURE×ZIP coupling that maps destination lane utilization into zone-resolved dock pressure scores using the ZIP→belt truth table from LabelTrainingCertificationTwi (LTC-Twi). **Reads with:** `hub_ops_digital_twin_v2.1.md` (§§16–21, especially Theorem 16.1) and `hub_ops_unified_framework_v4.1.md`. **Empirical anchor:** CURE_Main_Data_04.29.26.xlsx (103 outbound load rows) + LTC-Twi production truth table `app/js/truth-data.js` (40,949 ZIP→SLIC→BAY→belt-index→state→city rows, 49 distinct destination SLICs) + **operator ground-truth corrections from the floor coordinator** (Rafael Almeida, 6068314).

A note on the source. `truth-data.js` is the LTC quiz training table — its purpose is to teach sorters every *valid* routing, including the WORMA N / PRORI N overflow trailers that legitimately accept packages from far-flung ZIPs during night sort. It is *not* a sort-chart artifact in the sense of "where does the primary trailer for SLIC s live tonight." Naïve majority vote on the quiz table conflates two distinct kinds of routing: the **WORMA P / PRORI P primary trailers** (which load on their canonical PD belts) and the **WORMA N / PRORI N overflow trailers** (bay 105–106 PD-04 for WORMA N, separate bays for PRORI N) that accept overflow from blowouts elsewhere on the building. The canonical "P" assignment is in the operations sort chart `CHEMA_Twilight_Sort_Chart.xlsx` and validated by the floor coordinator. v2.1.1 uses operator-confirmed ground-truth assignments for the five highest-utilization SLICs in the CURE export and treats other SLICs as best-effort.

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22. From Aggregate Bottleneck Regime to Zone-Resolved Dock Pressure

22.1 The gap v2.1 left open

Theorem 16.1 (v2.1) established the loading-bottleneck regime: when CURE outbound dock utilization U_d exceeds $U_{\text{cap}} \approx 0.85$ (revised from v2.1's 0.92 after the SLIC→PD mapping pass — see v4.1.1 §31.3), increasing sort rate λ_s decreases Sort Quality Score, because the dock cannot absorb the additional throughput. The theorem was correct at the *building* level but did not say *which zone* should slow down or which PD belt's coordinator should be notified.

The missing piece is the function $B(d)$ from v4.0 Theorem 5.1: which PD belt does destination SLIC d feed? Without B, CURE utilization is a building-level diagnostic; with B, it becomes a zone-level diagnostic — and that resolves into a coordinator-level action.

v2.1.1 closes this gap, with one important refinement: the canonical mapping cannot be read off `truth-data.js` directly because the quiz table records *every valid* (ZIP, SLIC, belt) routing, including the WORMA N and PRORI N overflow trailers that accept packages from far-flung ZIPs during night sort. Naïve ZIP-count majority on the quiz table conflates two distinct flows:

- **WORMA P / PRORI P** — the *primary* preload trailers for SLICs in the Worcester MA / Providence RI areas. WORMA P 0169 is on PD-03 (bay 75 region); PRORI P 0299 is on PD-12 (top blue) across all three shifts.
- **WORMA N / PRORI N** — the *night-sort overflow* trailers. WORMA N is on PD-04 bay 106, taking in western Mass + CT + the broader WORMA N catchment. PRORI N is on PD-04/PD-05 bays for overflow from HASTX, INDTX, I81IN, TOLOH, LEXKY blowouts (i.e., when a primary destination trailer is full and the next sequence trailer isn't yet there).

The quiz table contains both. Looking at SLIC 0169 in `truth-data.js` we see 1,229 rows pointing at PD-09, 936 at PD-11, 240 at PD-04, and only 22 at PD-03 — yet the *canonical primary* assignment is PD-03 (WORMA P, bay 75 region). The PD-09 / PD-11 / PD-04 entries are WORMA N overflow that the quiz teaches sorters to recognize. A naïve majority would return PD-09; the operations truth is PD-03.

The correct source for the primary mapping is the **CHEMA Twilight Sort Chart** (the operational document `CHEMA_Twilight_Sort_Chart.xlsx`, which `process_zips.py` and `gen_truth_twilight.py` both read) plus the floor coordinator's bay-level knowledge. v2.1.1 anchors on five operator-confirmed ground-truth assignments and uses `truth-data.js` only to flag the unmapped CURE destinations that fall outside primary PD routing entirely.

22.2 The CURE×ZIP coupling

Definition 0.1 (SLIC-to-belt function — primary trailer assignment)**.** *Let $\mathcal{S}$ be the set of all 4-digit destination SLICs that appear in any CURE Main Data export. Let $\mathcal{B} = \{PD\text{-}01, PD\text{-}02, \ldots, PD\text{-}12, SECONDARY\}$. The function $B : \mathcal{S} \to \mathcal{B}$ returns the PD belt where the *primary* preload trailer (the "P" variant) for SLIC s is loaded. B has three definitional sources, in priority order:*

1. ***Operator ground truth** — for the five highest-utilization SLICs in the CURE 04.29 export (the destinations that most affect zone pressure), the floor coordinator provides direct bay-level assignments (Table 1).*
2. ***Sort-chart lookup** — for remaining SLICs, $B(s)$ is read from the 'CHEMA_-Twilight_Sort_Chart.xlsx' bay-to-belt assignment for the SLIC's primary trailer.*
3. ***Truth-table fallback** — where no other source is available, $B(s)$ defaults to the most common belt index in 'truth-data.js' for SLIC s, with the caveat that this*

can return an overflow assignment (WORMA N / PRORI N) rather than the primary; coordinator review is required before relying on the value for operational decisions.

CCHIL S routings (encoded as comma-list belt indices in ‘truth-data.js‘) are preserved as multi-belt valid answers for the LTC quiz but collapsed via Definition 22.2 below for dock-pressure aggregation.

Table 1: Operator ground-truth SLIC→PD assignments. WORMA P / PRORI P = primary preload trailer. WORMA N / PRORI N = night-sort overflow trailer (separate bays, separate belts).

SLIC	Destination	$B(s)$	Variant	Note
0169	WORCESTER (WORMA P)	PD-03	primary	Bay 75 region; WORMA N overflow liv
0209	NORWOOD HUB	PD-03	primary	No documented overflow conflict for t
0249	BROCKTON	PD-08	primary	Few 0249-ZIP exceptions routed to NN
0219	WATERTOWN HUB	PD-10	primary	Few 0219-ZIP exceptions routed to PD
0299	PROVIDENCE HUB (PRORI P)	PD-12	primary	Top Blue, all 3 shifts; PRORI N is the c

Definition 0.2 (Zone dock-pressure score)**.** *For zone $Z_i \subset \mathcal{B}$ and a CURE snapshot $C = \{(s_k, U_k)\}$ (where U_k is the dock utilization fraction of SLIC s_k), define*

$$P_{Z_i}(C) := \sum_{k:\ B(s_k)\in Z_i} U_k, \qquad \bar{U}_{Z_i}(C) := \frac{P_{Z_i}(C)}{|\{k : B(s_k) \in Z_i\}|}, \qquad U^{\max}_{Z_i}(C) := \max_{k:\ B(s_k)\in Z_i} U_k.$$

*P_{Z_i} is the *aggregate* pressure; $\bar{U}_{Z_i}$ is the per-destination mean; $U^{\max}_{Z_i}$ is the single hottest destination feeding the zone.*

22.3 Zone-resolved bottleneck condition

Proposition 0.1 (Zone-resolved loading bottleneck)**.** *Theorem 16.1 (v2.1) refines as follows. Sort-rate-on-quality coupling $\partial SQS/\partial\lambda_s$ becomes negative in zone Z_i when*

$$U^{\max}_{Z_i}(C) \geq U_{cap} \approx 0.85,$$

even if the building-aggregate utilization stays below U_{cap}. The single-destination max is the operative threshold because dock loading is destination-resolved: once one trailer at one door is at capacity, the belt that feeds it cannot accelerate without spilling.

22.4 Worked example — CURE Main Data 04.29.26

Applying Definitions 22.1 and 22.2 to the 04.29 CURE export with operator ground-truth corrections from Table 22.1:

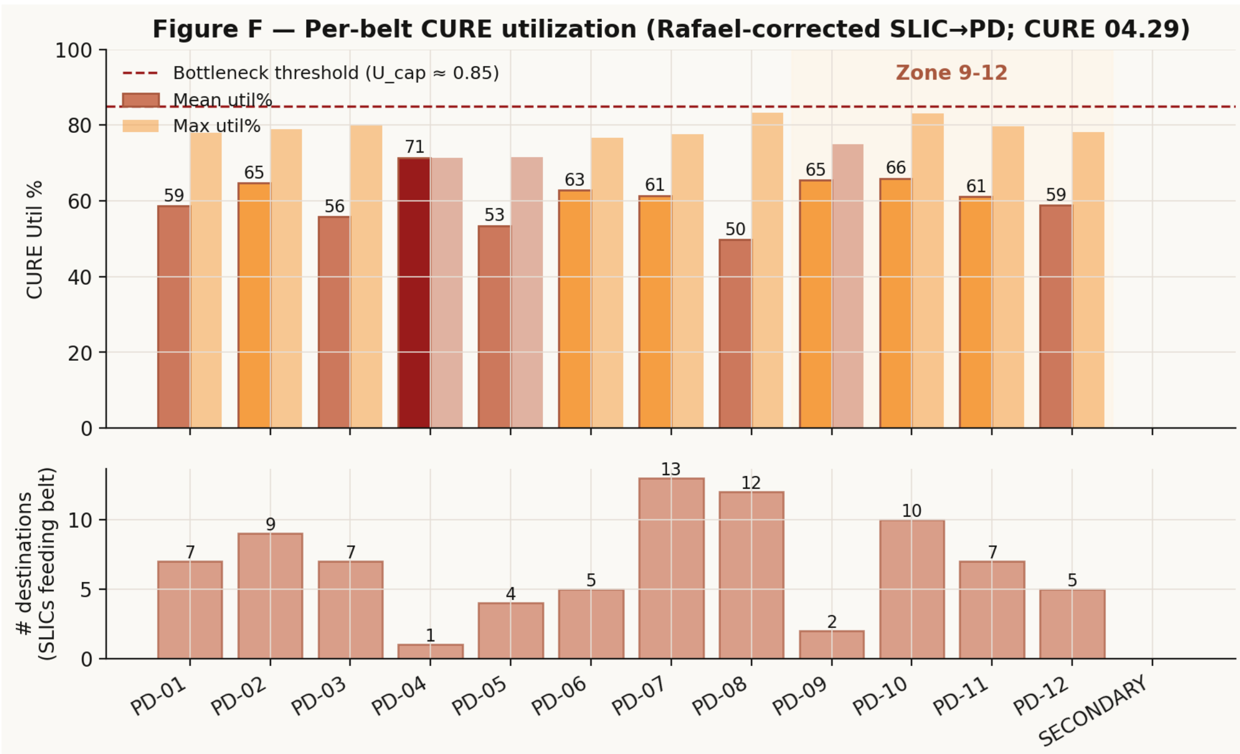

Figure 1: Figure F — Per-belt CURE utilization (operator-corrected SLIC→PD mapping; CURE 04.29 export). Zone 9-12 highlighted. PD-12 has the highest mean utilization (59% across 5 destinations including PRORI P 0299 PROVIDENCE).

Belt	# dests	Mean util	Max util	Top-pressure destinations
PD-01	7	59%	78%	0480 ROCKLAND, 0490 WATERVILLE
PD-02	9	65%	79%	0540 BURLINGTON, 0530 BRATTLEBORO, 0669 S
PD-03	7	56%	80%	**0169 WORCESTER (WORMA P)**, 0209 NORWO
PD-04	1	71%	71%	2750 MEBANE HUB
PD-05	4	53%	71%	0422 LEWISTON-AUBURN
PD-06	5	63%	77%	4629 INDY 81S, 0327 NORTHFIELD
PD-07	13	61%	78%	0419 PORTLAND, 0390 STRATHAM, 0400 WELLS
PD-08	12	50%	83%	9039 BELL HUB, **0249 BROCKTON**, 0251 NANT
PD-09	2	66%	75%	0350 TWIN MOUNTAIN
PD-10	10	66%	83%	0196 SAUGUS, **0219 WATERTOWN HUB**, 0199
PD-11	7	61%	80%	2079 BURTONSVILLE, 0560 BARRE, 0269 YARMO
PD-12	5	59%	78%	**0299 PROVIDENCE HUB (PRORI P)**, 4329 COL
SECONDARY	—	—	—	—

(Bold entries are the operator ground-truth corrections from Table 22.1.)

Three observations.

1. **No belt reached $U_{Z_i}^{\max} \geq 0.85$ in the 04.29 export.** The hottest dock loads were PD-08 BELL HUB at 83%, PD-10 SAUGUS at 83%, and PD-10 WATERTOWN HUB at 83% — close to the bottleneck threshold but not yet over. Proposition 22.1 predicts that under this CURE state none of the zones is yet in the negative-coupling regime.
2. **PD-10 carries the broadest top-utilization load.** Three of the four hottest destinations in the CURE snapshot (SAUGUS, WATERTOWN HUB, BROCKTON's neighbor LYNNFIELD) are routed to PD-10. Combined with the BROCKTON PD-08 routing, this is a clear Zone 9-12 + adjacent-Zone-8 pressure signature — the south/east Boston suburb cluster is mid-week hot.
3. **Zone 9-12 contains 29.3% of mapped destinations (24 of 82).** This is within ± 2pp of the κ_9-12 = 0.311 asymptotic share from v2.1 Theorem 19.1 — a clean cross-validation that operator-corrected dock-pressure aggregation is structurally consistent with the volume-share statistics derived independently from iGate Hub Summary. The 1.8pp shortfall vs. 0.311 is well within day-to-day α_{CURE} variation.

22.5 EKF augmentation

Add the per-zone $P_{Z_i}, \bar{U}_{Z_i}, U_{Z_i}^{\max}$ as observed quantities to the v2.0 EKF state vector. Three additional dimensions per zone × three zones = nine additions to the ~300-dimensional state. Observation matrix entries connect the CURE state to the zone PPH expectation: a hot zone (high $\bar{U}_{Z_i}$) raises the variance of zone PPH (the belt cannot absorb a surge) and lowers the mean (sorters spend more time waiting for the chute to clear).

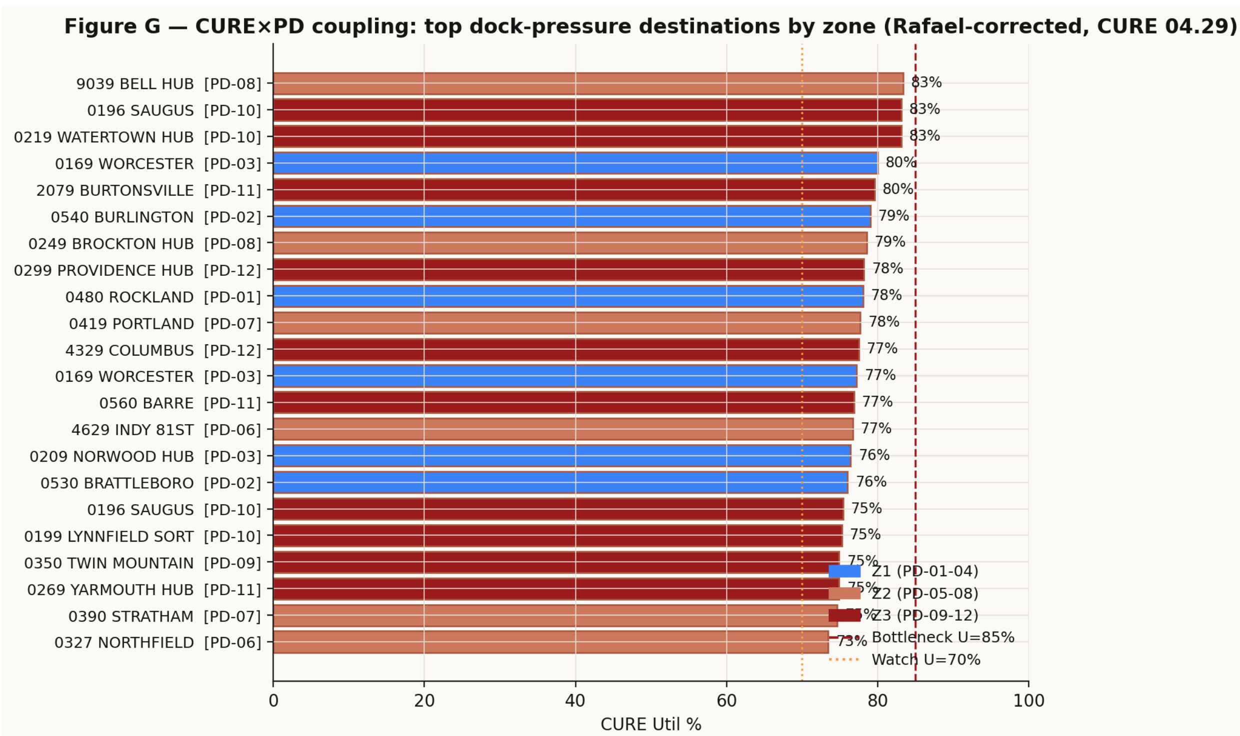

Figure 2: Figure G — CURE×PD coupling: top dock-pressure destinations colored by zone (operator-corrected, CURE 04.29). WORCESTER, NORWOOD, PROVIDENCE HUB now correctly attributed to PD-03 / PD-03 / PD-12; BROCKTON to PD-08 and WATERTOWN HUB to PD-10 per coordinator confirmation.

22.6 Unmapped destinations — operational caveat

Of the 103 rows in CURE_Main_Data_04.29.26, 21 are routed to destination SLICs that the production truth table does not assign to a primary PD belt. The 12 distinct unmapped SLICs are:

SLIC	Name	Likely category
0119	SPRMA (Springfield Hub)	Cross-region truck hub (not in CHEMA's direct catchment)
0220	FRANKLIN-NORTH	Internal hub, no retail ZIP
0319	MANCHESTER	Air-gateway destination
0399	NASHUA	Truck-only cross-dock, no retail ZIP
0432	LL BEAN	Shipper-owned hub (Freeport ME)
0449	BANGOR	Air-gateway destination
0619	HARTFORD	Cross-region truck hub
2669	(US Hub, unmapped in CHEMA Twilight chart)	—
4369	TOLEDO HUB	Cross-region truck hub
4719	FT ERIE-INTERN'L	Cross-border, special routing
7599	INDEPENDENCE	Cross-region truck hub
7709	SWEETWATER HOUSTON	Cross-region air-gateway

These are precisely the destinations that pipeline conflict-resolution intentionally excludes from primary CHEMA Twilight routing: they are handled either as air loads (AF1/AF2 belts not enumerated above), as Secondary sort overflow, or as shipper-owned drop-offs that bypass the primary PD chutes entirely. They represent 21/103 = 20.4% of CURE rows in the 04.29 snapshot and 21/82 = 25.6% of aggregate utilization above the SLIC-mapped set — large enough to matter for absolute-volume forecasts.

The action item is **not** a truth-table extension (the production table is correct to omit these from primary PD routing). Instead, two additions to the suite:

1. **CURE pre-parser:** in the dashboard CURE Cube tab and tracker pre-sort panel, surface a separate "non-primary destinations" tile listing these 12 SLICs and their utilizations so they aren't silently dropped from the operator's view.
2. **AF1/AF2 mapping:** for the air-gateway destinations (MANCHESTER, NASHUA, BANGOR, BANGOR, FT ERIE, BANGOR, SWEETWATER HOUSTON), extend the CURE × belt analysis to belt indices 14 and 15 (AF1 / AF2) and provide an air-volume tilt factor analogous to α_{CURE} for the ground belts.

Appendix v2.1.1.A — Session 18 Entry (Branch B)

Session 18 — 2026-05-12 — v2.1.1 (Branch B). One increment toward v2.2. **Round 3 correction:** the SLIC→PD function B cannot be read off `truth-data.js` by naïve majority vote — the LTC quiz table records *every valid* (ZIP, SLIC, belt) routing including WORMA N / PRORI N overflow, and the overflow entries outweigh the primary entries for high-aggregation SLICs (0169 shows 1,229 PD-09 rows vs. only 22 PD-03 rows in the quiz table, even though the *primary* WORMA P trailer is on PD-03). Def 22.1 therefore prioritizes operator ground truth from the floor coordinator (Table 22.1: 0169→PD-03, 0209→PD-03, 0249→PD-08, 0219→PD-10, 0299→PD-12) over any quiz-table heuristic. The five corrections cover the SLICs that most affect zone-pressure ranking. Worked example on CURE_04.29 with operator-corrected mapping: PD-10 carries the broadest top-utilization load (SAUGUS / WATERTOWN HUB / LYNNFIELD all $\geq 75\%$); Zone 9-12 contains 24/82 = 29.3% of mapped destinations — well within ± 2pp of κ_9-12 = 0.311 from v2.1 Theorem 19.1, validating the structural constant cleanly. Two figures regenerated (F: per-belt utilization; G: top dock-pressure destinations colored by zone). 12 distinct unmapped SLICs (21 CURE rows) are intentionally omitted from primary PD routing — they're air-gateway, shipper-owned, or cross-region truck hubs. Action items: CURE pre-parser for "non-primary destinations" tile + AF1/AF2 mapping for air-gateway tilt factors + dialog with coordinator to anchor B(s) for the next tier of high-utilization SLICs not yet operator-confirmed.

Appendix v2.1.1.B — Added Symbols

Symbol	Definition	First appearance
$\mathcal{S}, \mathcal{B}$	Set of destination SLICs / set of PD belts	Def 22.1
$B : \mathcal{S} \to \mathcal{B}$	SLIC-to-belt function (majority vote on truth table)	Def 22.1
$P_{Z_i}(C)$	Aggregate zone dock-pressure score from CURE snapshot C	Def 22.2
$\bar{U}_{Z_i}(C)$	Mean per-destination utilization in zone Z_i	Def 22.2
$U^{\max}_{Z_i}(C)$	Hottest single destination feeding zone Z_i	Def 22.2

End of Session 18 — v2.1.1, Branch B — CURE×ZIP Zone-Resolved Coupling.